Catching Meter Fraud Before It Costs Millions

How machine learning helps STEG find fraud and reduce its losses

Tim Pfeiffer · Irem Baykal · Dhanya Rajendran

Agenda

Part 1 · Business case

Who we are, what STEG loses to meter fraud, and why we frame it as a ranking task.

Slides 3 to 5

Part 2 · EDA and ML

What the data shows, which signals stand out, how we compared the models, and how well the final model works on unseen test data.

Slides 6 to 13

Part 3 · Idea and action

What STEG could gain, how a dashboard and an inspector app would work, the limits of our data, and our recommendation.

Slides 14 to 19

Who we are and what STEG faces

Three data analysts, one fraud problem: meter tampering at STEG

Who we are

We are three data analysts, the **Fraud Busters.**

Tim Pfeiffer
Irem Baykal
Dhanya Rajendran



We built a fraud risk system for STEG.

STEG is Tunisia's state electricity and gas utility. Some clients tamper with their meters, and inspectors can only visit a few homes.

135,493

clients, with 4.5 million invoices*

69% / 31%

of invoices are electricity / gas

7,566

fraud clients, only 5.6% of all

* Period 2005 - 2019

STEG loses ...

200 MILLION

dinars

to meter tampering*.

* Period 2005 - 2019

Our idea: a combined fraud system that helps STEG to discover and fight fraud



Starting point: the data* we looked at for training

What we have, and what we want to find

135,493 clients

4.5 million bills

22 features

Goal: find fraud

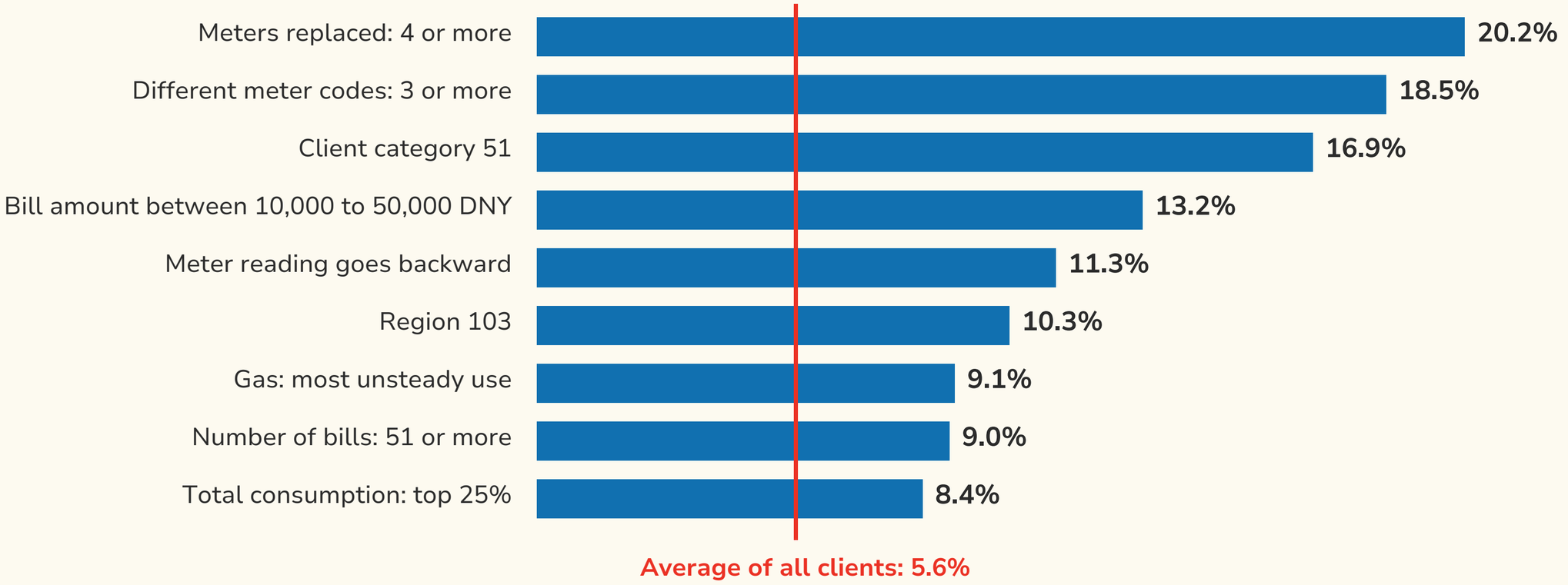
Fraud is rare and hidden in millions of bills

5.6% of clients are fraud (imbalanced)

* Period 2005 - 2019

Which features go with fraud?

Share of fraud amongst clients by feature.



Three signals up close

Fraud share rises with meter changes, unsteady gas use and backward readings

4 or more meters replaced

20.2%

1 meter: 2.1%,
2 meter: 6.2%,
3 meter: 14.8%.

Unsteady gas use

9.1%

fraud for clients whose gas use jumps the most, vs 5.1% for the steadiest.

Meter reading backward

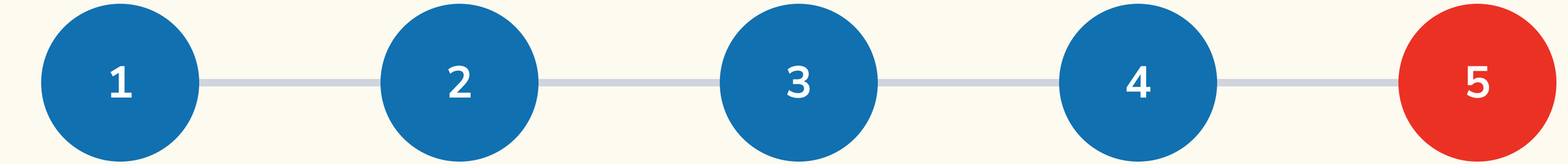
11.3%

fraud vs 5.5% for all other clients.
Rare: only 1,816 clients.

Next Step: Top features condensed in our EDA will be used for setting up our ML-Model

ML pipeline: from raw bills to a ranked list

Early 80/20 split, learned steps fitted on train only, 5-fold stratified CV



Clean data (same rules for train and test)

Feature engineering: one row per client

Compare 4 models (5-fold CV)

Tune LightGBM (random search)

Error analysis, then new features

LightGBM ranked best, then tuning helped

MODEL	ROC-AUC
Logistic regression (baseline)	0.757
Random Forest	0.743
XGBoost	0.743
LightGBM, default	0.767
LightGBM, tuned (24 features)	**0.774**

FINAL MODEL BEFORE ERROR ANALYSIS

Features from error analysis drove the gain

Error analysis: the model missed low-consumption fraud, long histories and young clients

01

Meter features

How often a client's meter was replaced.

02

Time features

Client age, history length and gaps between bills.

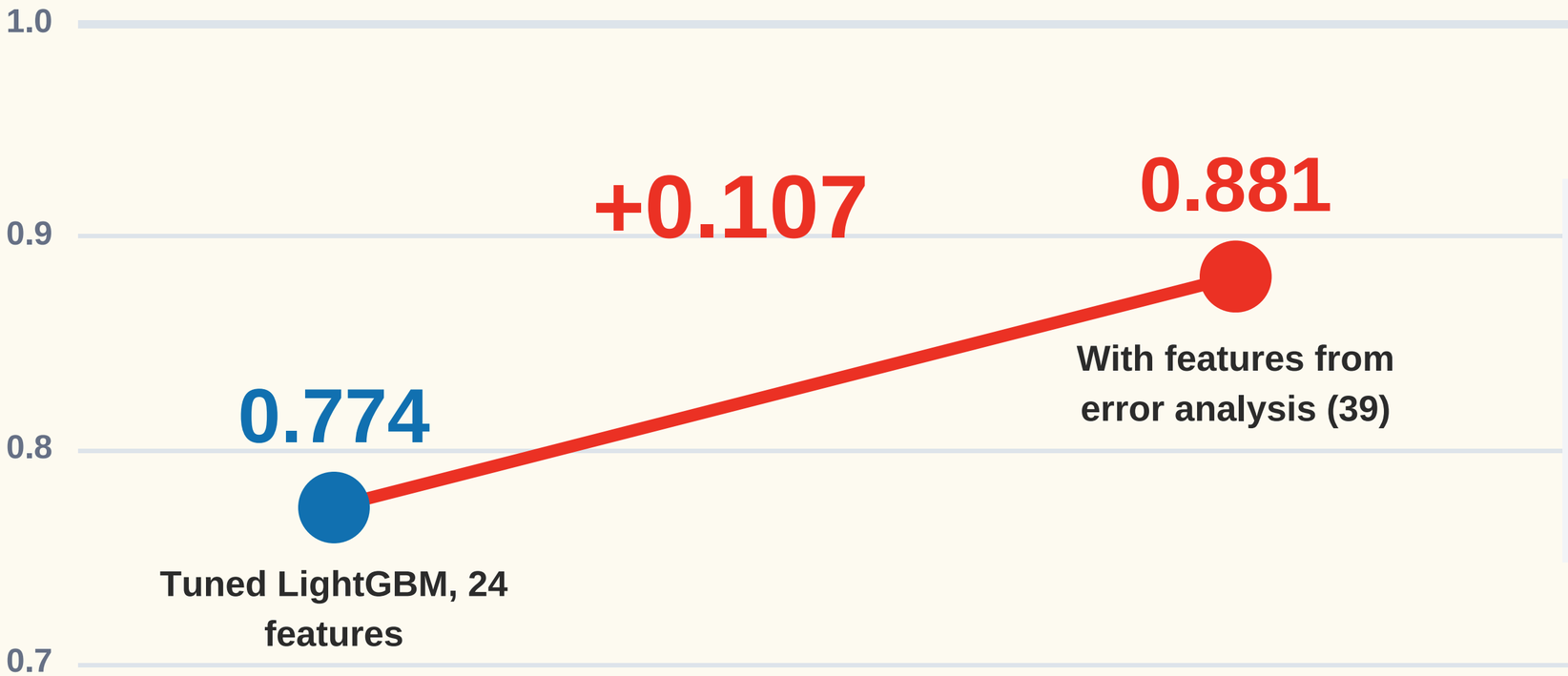
03

Diversity features

How varied the meter setup is.

Error analysis lifted the score from 0.774 to 0.881

The new features lifted the score a lot more than tuning did



47% fewer mistakes

Fraud ranked below honest: 23 in 100 before, 12 in 100 now

Test score: 0.874 on unseen clients

Test set: clients whose fraud labels the model never saw during training or tuning

0.874

ROC-AUC ON THE TEST SET

87%

FRAUD RANKED ABOVE HONEST CLIENTS

PRECISION	RECALL
0.367	0.472

Trained once on all labelled clients, with no tuning on the test set.

Inspecting customers: 6x more fraud caught

Share of all fraud found



IF STEG INSPECTS 10% OF CLIENTS

1 in 3

finds fraud (instead of 1 in 18)

Estimated recovered losses: about
94 million TND per year

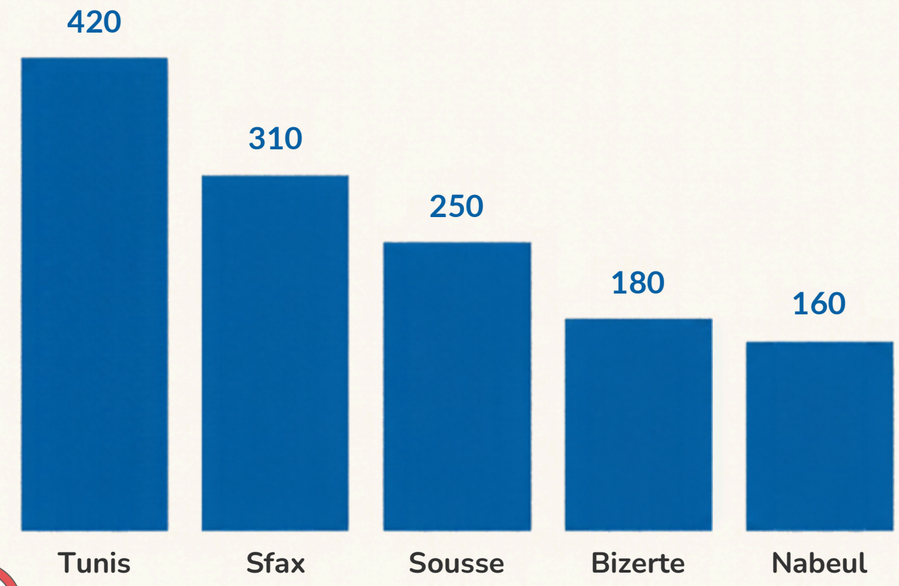
Managers see where to inspect each month

Prototype: monthly manager view with regional workload and a top-10 risk list

PART 3 - IDEA & ACTION

EXAMPLE DATA - FICTIONAL

Clients to inspect per region



34%

EXPECTED HIT RATE

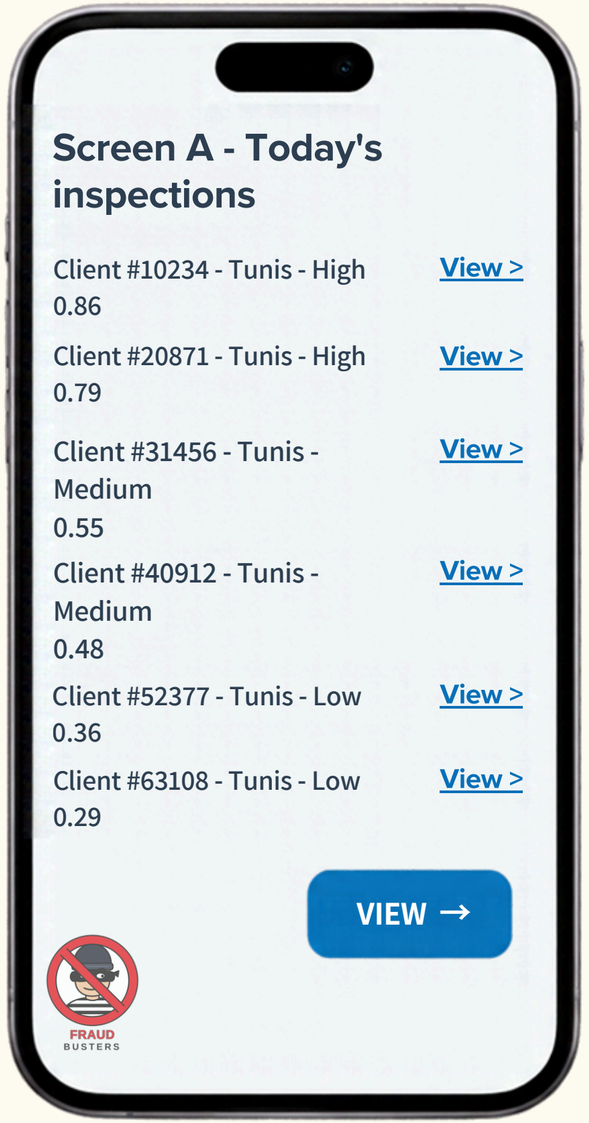
TOP 10 RISKIEST CLIENTS

#1876	Tunis	0.93
#24510	Sfax	0.91
#37102	Sousse	0.90
#41892	Bizerte	0.89
#55621	Nabeul	0.88
#66045	Gabes	0.87
#10224	Tunis	0.86
#27733	Sfax	0.84
#30130	Sousse	0.83
#40027	Bizerte	0.82

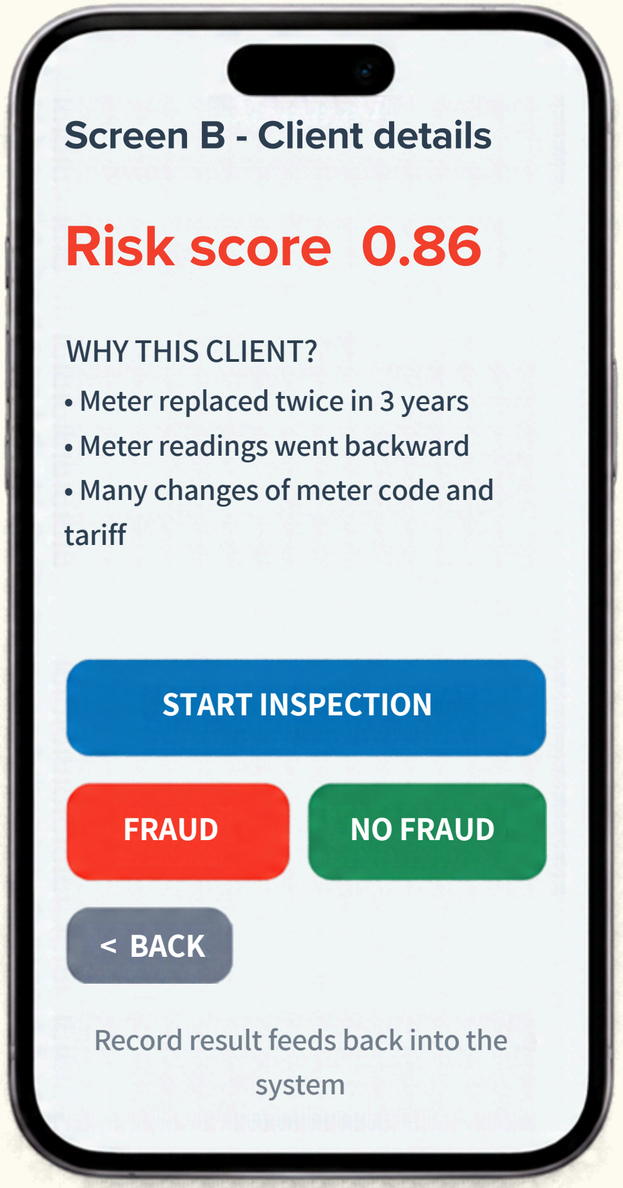
FRAUD BUSTERS

Inspectors get a ranked list on their phone

PROTOTYPE · EXAMPLE DATA · FICTIONAL



PROTOTYPE · EXAMPLE DATA · FICTIONAL



Limitations and what we need to go live

1

Label bias

Labels only show detected fraud. Pilot inspection results give unbiased feedback and improve the model.

2

Possible leakage

n_counters (meter replacements) and days_since_last_bill (inactive accounts) may reflect what STEG did after finding fraud: check with STEG's records.

3

Fraud without traces

Some fraud (small, stable, young accounts) leaves no pattern in the bills, so inspectors' experience stays important.

Recommendations: pilot in one region

Three steps to validate the model on STEG's own inspections

01

Start a pilot

In one region. Compare the results with today's inspections.

02

Feed it back

Use the risk list every month and feed the inspection results back, so the system keeps learning.

03

Share records

Share meter-replacement and inspection records with us to confirm the strongest signals.

Fewer wasted visits. More fraud found. Lower losses.

Thank you.
Questions?

Tim Pfeiffer · Irem Baykal · Dhanya Rajendran

Appendix: links

Where to find the code, the data and the competition

Project repository

github.com/IremLeyla/ds-ml-project

Zindi competition

zindi.africa/competitions/fraud-detection-in-electricity-and-gas-consumption-challenge

Zindi data page (the data itself is not in our repo)

zindi.africa/competitions/fraud-detection-in-electricity-and-gas-consumption-challenge/data

Published solutions we compared with

github.com/imgremlin/Fraud-Detection-by-Zindi

github.com/AmirFARES/Tunisia_Energy_Fraud_Detection_STEG

Start in the repo with 00_overview.md: every notebook in reading order, from data ingestion to the final model.